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AkiNet: A Physics-Informed AI for Wave Extraction From Noise

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Key Points:

- AkiNet, a Physics-Informed Neural Network framework, solves the nonlinear inverse problem of surface wave dispersion from noise correlation functions
- AkiNet performs the inversion in an end-to-end process, requiring neither a preliminary feature-extraction step nor an initial model guess
- AkiNet employs unsupervised learning, embedding physical constraints directly into its network architecture and loss function

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract High-resolution seismic models of the Earth's lithosphere are critical for understanding its structure and evolution, yet current global models lack the details that can be provided by ambient noise data. A primary bottleneck is reliably extracting phase velocities from the vast, often noisy data sets produced by ambient noise cross-correlations (noise correlation functions). To address this, we introduce *AkiNet*, a Physics-Informed Neural Network designed as a “zero-shot” solver for this difficult inverse problem. Unlike supervised learning approaches, AkiNet operates without pre-training or labeled data by directly embedding the governing physics of wave propagation (Aki's theoretical framework) into its loss function. When compared against a modern waveform-fitting algorithm, AkiNet yields more reliable dispersion estimates, particularly for Love wave data with low signal-to-noise ratios. AkiNet provides a robust and scalable tool that presents a feasible pathway toward the construction of a global, high-resolution, and noise-derived dispersion model.

Plain Language Summary To see inside the Earth, seismologists study how waves from subtle vibrations, known as *ambient seismic noise*, travel through it. Getting useful information from these seismic data is like trying to pinpoint your location on a map using a single, fuzzy GPS signal. Our new tool, *AkiNet*, solves this problem by acting like an advanced GPS. First, it uses the fuzzy signal (the seismic data), but critically, it also has a complete map of the road network (the laws of wave physics). By insisting the final location must be on a road, AkiNet can find a precise answer even when the original signal is poor. We make our tool even smarter by giving it extra hints, such as preferring simpler, smoother routes (a technique called regularization) and using general knowledge of Earth's structure to narrow down the search area. This approach allows scientists to create more reliable and detailed maps of Earth's interior, paving the way for a better understanding of our planet.

1. Introduction

Seismological models of the lithosphere are fundamental to understanding Earth's structure, composition, and geodynamic evolution (Hacker et al., 2015; Rudnick & Fountain, 1995; Sammon & McDonough, 2022; Wipperfurth et al., 2020). A key pathway to improving these models, particularly their vertical resolution, is through the assimilation of surface wave dispersion data derived from ambient noise interferometry. While this approach has been successfully applied at continental scales (Olugboji et al., 2024), prominent global models such as CRUST1.0 (Mooney et al., 1998), Litho1.0 (Pasyanos et al., 2014), and ECM1 (Mooney et al., 2023) have not yet assimilated such noise-derived dispersion measurements. Consequently, their resolution remains limited within the continental crust and oceanic upper mantle. Two primary obstacles hinder this global integration: (a) the immense computational cost of calculating noise cross-correlations on a global scale and (b) the methodological difficulty of reliably inverting these correlations for phase velocity, given their often low signal-to-noise ratios. This paper introduces a novel approach, using Physics-Informed Neural Networks (PINNs), to tackle the second challenge.

Methodologies for inverting noise correlation functions (NCFs) for phase velocity dispersion broadly fall into two categories. The first includes traditional physics-based algorithms, which either fit the zero-crossings of the NCF (Ekström et al., 2009) or perform a linearized or (Menke & Jin, 2015) nonlinear (Hawkins & Sambridge, 2019) waveform-fitting of the entire NCF. These methods are grounded in physical theory, but can be sensitive to noise and require careful parameterization. The second category leverages supervised machine learning, which has shown success in other geophysical tasks like earthquake detection (Mousavi & Beroza, 2022; Perol et al., 2018)

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and phase picking (Mousavi et al., 2020; Ross et al., 2019). However, its application to dispersion estimation has been limited to “black-box” models that depend on a vast quantity of high-quality labeled data for training, which are often unavailable (Dai et al., 2021; Dong et al., 2021; Song et al., 2021; Zhang et al., 2020). To overcome these respective limitations, we introduce a third approach, AkiNet, which is built upon the Physics-Informed Neural Network (PINN) framework.

PINNs are a class of deep learning models originally conceived for solving partial differential equations (PDEs) (Alkhalifah et al., 2021; Cheng & Alkhalifah, 2025; Raissi et al., 2019). Recently, the term has broadened to describe any neural network that embeds prior physical knowledge—algebraic equations, conservation laws, or derivative constraints—directly into its architecture or loss function to guide the solution (Cuomo et al., 2022; Faroughi et al., 2024; Karniadakis et al., 2021). A prominent geophysical example of this modern paradigm is EikoNet (Smith et al., 2020), where the Eikonal equation is not solved in the forward pass but is instead enforced as a physical constraint within the loss function via automatic differentiation. This approach is also extended to HypoSVI, which solves the complex inverse problem of hypocenter location (Smith et al., 2022). AkiNet’s architecture is built under this same principle. By encoding known physics directly into the network’s loss function, this framework can circumvent the reliance on massive labeled data sets and exhibit better generalization. In geophysics, this paradigm has been used to develop fast surrogate models for forward problems, (Smith et al., 2020), and more recently, as powerful tools for solving ill-posed inverse problems like full-waveform inversion (Haghighat et al., 2021; Rasht-Behesht et al., 2022).

In this paper, we introduce AkiNet, a novel inversion tool that exemplifies this modern PINN paradigm. Named in honor of Aki (1957), our approach directly embeds his theoretical framework—the representation of the NCF with Bessel functions (Nishida, 2017)—within the network’s loss function. This design allows AkiNet to operate in an unsupervised fashion to directly address the core challenge outlined in this paper: the robust extraction of phase velocity dispersion from NCFs with disparate signal-to-noise ratios. We use the term “unsupervised” to emphasize to the geoscience audience that our method requires no pre-labeled data and operates on a single instance at a time, in direct contrast to supervised methods. This approach shares principles with “self-supervised” learning, but we find “unsupervised” provides a clearer distinction in this context. Sections 2 and 3 describe data preparation as well as the foundational theory and AkiNet’s architecture. Section 4 presents inversion examples and a systematic comparison against a traditional method using a large data set. Finally, Section 5 discusses the implications of our results and potential avenues for future research.

2. Data and Pre-Processing

The input data used in the AkiNet framework is the Noise Cross-correlation Function (NCF), which represents the coherent surface wave energy traveling between a pair of seismic stations. For this study, we use the publicly available data set of NCFs from across the African continent (Olugboji & Xue, 2022). While the full processing details are available in Olugboji and Xue (2022), we summarize the key steps for completeness.

The NCFs, are computed in the frequency domain using the spectral coherence method (Jiang & Denolle, 2020; Prieto et al., 2009):

$$\rho(f, r) = \frac{U_A(f)U_B^*(f)}{|U_A(f)||U_B(f)|} \quad (1)$$

where $U_A(f)$ and $U_B(f)$ are the Fourier transforms of the vertical or transverse component seismograms from stations A and B, respectively, and the asterisk denotes the complex conjugate. The raw seismograms were downsampled to 1 Hz after instrument response removal to obtain ground displacement. Following standard practice, Rayleigh wave NCFs are generated from vertical-vertical correlations, while Love wave NCFs are from transverse-transverse correlations (Haney et al., 2012).

A key distinction of our methodology lies in how we use this NCF. Many deep learning applications first perform a “feature extraction” step, transforming the 1D time-domain NCF into a 2D image, typically via Frequency-Time Analysis (FTAN) or an F-J Bessel transform (Dong et al., 2021; Wu et al., 2020). The dispersion problem is then recast as an image segmentation task to pick a curve from this 2D representation (Dong et al., 2021; Zhang et al., 2020). In contrast, AkiNet completely bypasses this feature extraction step. We model the 1D NCF

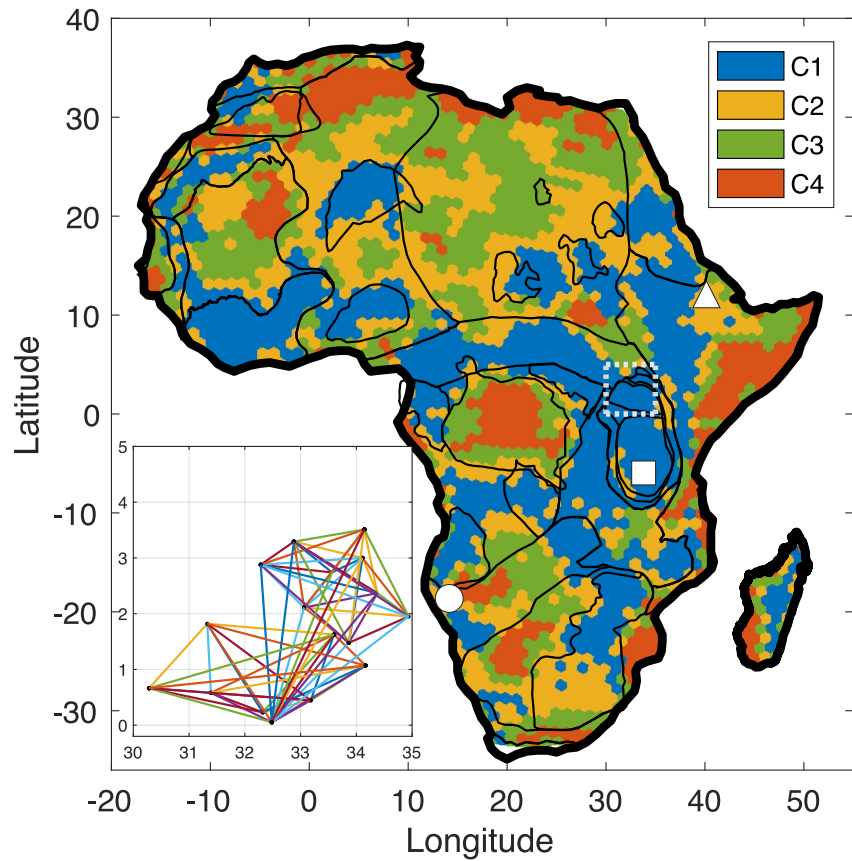


Figure 1. Map of Africa showing four distinct crustal types (C1–C4) from Olugboji et al. (2024). The dashed white box indicates the $5^\circ \times 5^\circ$ study area within the Ugandan Craton. The white square marker denotes the location for station pair XD-RUNG_XD-MTAN; the white circle marker denotes 6A-WP22_6A-WP11; and the white triangle marker denotes 2H-ASYE_7C-GOR. Inset: Locations of the 17 seismic stations (black dots) and the interstation paths analyzed in the main text. In our generalization tests, we draw 539 station pairs from all the different crustal types. Line colors are for visual distinction only.

spectrum directly, framing the task as a direct waveform-fitting inverse problem based on Aki's SPAC theory (Menke & Jin, 2015; Tsai & Moschetti, 2010).

To validate our method, we use NCFs from four geologically distinct regions across Africa (Figure 1), chosen to provide a diverse and challenging test set based on the tectonic classifications in Olugboji et al. (2024). Specific, challenging examples are drawn from station pairs located within the Ugandan Craton, while our large-scale benchmark tests utilize 539 NCFs sampled from across these different tectonic domains.

3. Method

3.1. The Forward Problem: Modeling the NCF Spectrum

The goal of AkiNet is to solve the nonlinear inverse problem: retrieving the frequency-dependent phase velocity, $c(f)$, from an observed NCF. f is the frequency. The forward problem, which mathematically connects $c(f)$ to the NCF, is based on Aki's theoretical framework (Aki, 1957). This framework models the NCF spectrum, $\rho(f, r)$, as a linear combination of Bessel functions. r is the inter-station distance. In this paper, we use $c(f)$ and $\rho(f, r)$ to denote phase velocity and NCF generically. For Rayleigh waves, the NCF spectrum is vertical-vertical (ZZ) cross-correlations, and is described by the zeroth-order Bessel function of the first kind, J_0 :

$$\rho(f, r) = AJ_0\left(\frac{2\pi fr}{c(f)}\right). \quad (2)$$

For Love waves, the transverse-transverse (TT) cross-correlation is used, and the expression involves both the zeroth- and second-order Bessel functions (Nishida et al., 2024):

$$\rho(f, r) = A \left[J_0 \left(\frac{2\pi f r}{c(f)} \right) - J_2 \left(\frac{2\pi f r}{c(f)} \right) \right] / 2. \quad (3)$$

In these equations, A is a frequency-dependent amplitude term related to the strength and distribution of ambient noise sources. Our objective is to invert these equations for $c(f)$ given an observed NCF. $c(f)$ within the Bessel functions makes this inversion problem nonlinear, bringing more challenges.

The remainder of this section details our methodology for solving this inverse problem. We begin by describing the pre-processing step used to account for the unknown amplitude term A through NCF normalization (Section 3.2). We then introduce the architecture of AkiNet, the PINN that performs the nonlinear inversion (Section 3.3). To place our method in context, we compare the AkiNet framework with traditional inversion algorithms (Section 3.4) and conclude by detailing our approach for quantifying the uncertainty of the final dispersion curve (Section 3.5).

3.2. Amplitude Normalization: Isolating Phase

A successful inversion for $c(f)$ requires isolating the phase information in the NCF, as it contains the most direct signature of Earth structure. A common challenge, however, arises from the uneven distribution of ambient noise sources. This causes the amplitude of the observed NCF, ρ^{obs} , to deviate from the theoretical Bessel function model presented in Equations 2 and 3 (Hawkins & Sambridge, 2019). If not corrected, a waveform-fitting inversion will attempt to match both phase and amplitude simultaneously, potentially compromising the accuracy of the phase fit. The primary goal is therefore to decouple the fitting of amplitude from that of phase.

Previous work addresses this by modifying the forward predictions during the inversion. For example, Hawkins and Sambridge (2019) first compute an envelope of the observed NCF and then use it to scale the amplitude of the predicted NCF in each iteration. In AkiNet, we achieve the same essential goal with a different implementation: we normalize the observed NCF in a single pre-processing step so its amplitude spectrum is consistent with the theoretical model (i.e., where $A = 1$). While the implementations differ, both strategies effectively remove amplitude as a variable in the misfit calculation, thereby forcing the optimization to focus on the structurally sensitive phase information.

This pre-processing approach enhances convergence stability for our neural network. For a PINN that starts from a random initialization (i.e., without a well-informed prior for $c(f)$), iteratively correcting for amplitude within the training loop can lead the optimization toward undesirable local minima. Normalizing the data beforehand presents a simpler, better-posed target for the network to fit. Our normalization uses the analytical signal, computed via the Hilbert transform $\mathcal{H}(\cdot)$, to define the envelope $\mathcal{E}(\cdot)$ of the observed NCF:

$$\mathcal{E}(\rho^{\text{obs}}(f)) = |(\rho^{\text{obs}}(f)) + i\mathcal{H}(\rho^{\text{obs}}(f))|. \quad (4)$$

We then use this observed envelope, along with the theoretical envelope of the Bessel function, to normalize the NCF:

$$\rho^{\text{norm}} = \rho^{\text{obs}} \times \frac{\mathcal{E}(J_0(r, f))}{\mathcal{E}(\rho^{\text{obs}}(f))}. \quad (5)$$

This step not only stabilizes the inversion but also reduces computational overhead, as envelopes are not calculated repeatedly during training. An example of this normalization is shown in Figure S1 in Supporting Information S1.

3.3. The AkiNet Architecture and Training

Conceptually, AkiNet is a PINN composed of three core components: (a) a neural network that maps the input variable (frequency) to the desired solution (phase velocity), (b) a physics-informed loss function that measures the misfit between the network's prediction and the observed data, and (c) a training mechanism that uses

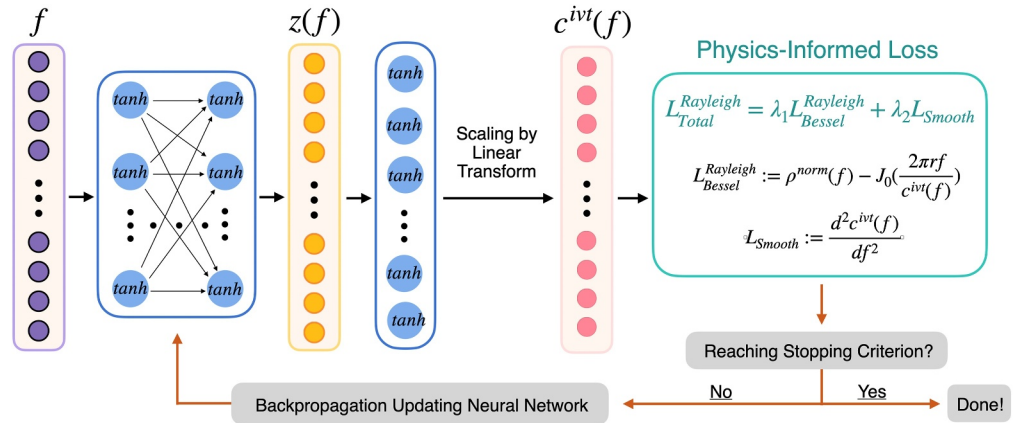


Figure 2. Illustration of AkiNet structure and loss function for Rayleigh Wave.

backpropagation to optimize the network's parameters by minimizing the loss. While not solving a canonical PDE in its forward pass, AkiNet's architecture aligns with the modern PINN paradigm exemplified by geophysical deep learning models like EikoNet (Smith et al., 2020) and HypoSVI (Smith et al., 2022). These methods leverage the power of automatic differentiation during back-propagation to enforce physical constraints within the loss function. A key advantage of this approach is that it provides effortless access to higher-order derivatives of the network output with respect to its parameters. This capability is highly valuable for complex, downstream inverse problems, such as computing the sensitivity kernels needed for large-scale path-averaged model updates (Hawkins & Sambridge, 2019; Menke & Jin, 2015).

3.3.1. The Feed-Forward Neural Network

The core of AkiNet is a feed-forward neural network that directly parametrizes the phase velocity function $\mathbf{c}(f)$ (Figure 2). The network's input is a vector, $\mathbf{f}$, containing N discrete frequencies. This vector is processed through four fully connected hidden layers. Based on experimentation, we find that 40 neurons per layer with a $\tanh$ activation function provides a sufficient balance between model expressivity and resistance to overfitting.

A key architectural feature of AkiNet is a final transformation layer that constrains the network's output to a physically plausible range. The network's last hidden layer produces an intermediate vector, $\mathbf{z}(f)$, which is passed through a $\tanh$ activation function. This maps the output to a normalized vector, $\hat{\mathbf{z}} = \tanh(\mathbf{z})$, where each element is bounded within $[-1, 1]$. This normalized vector is then scaled and shifted by a linear transformation to yield the final inverted phase velocity, $\mathbf{c}^{\text{ivt}}$:

$$\mathbf{c}^{\text{ivt}}(f) = \mathbf{W}\hat{\mathbf{z}}(f) + \mathbf{b}(f)$$

This transformation is designed to act as a physically informed bounding box. The term $\mathbf{b}(f)$ serves as a simplified, linear reference velocity curve. While not a direct fit to a specific Earth model, its linear form enforces the expected physical trend of decreasing velocity with increasing frequency and ensures that more complex reference models (e.g., AK135) are contained within the final solution space. The specific parameters that define separate reference curves for Rayleigh ($\mathbf{b}_R$) and Love ($\mathbf{b}_L$) waves are provided in Text S1 in Supporting Information S1.

The scaling matrix, $\mathbf{W}$, is a diagonal matrix that defines the maximum allowable perturbation around this reference velocity curve. We use a constant diagonal value (e.g., 0.8), a hyperparameter chosen to be generous enough to span all plausible velocities required by the data. This effectively creates a solution corridor of ± 0.8 km/s around the reference line. Together, these components ensure the final dispersion curve remains within a permissible range (see Section 4.2 for details). This architecture embeds prior geophysical knowledge directly into the network, acting as a strong regularizer that guides the inversion toward physically meaningful solutions without being overly restrictive.

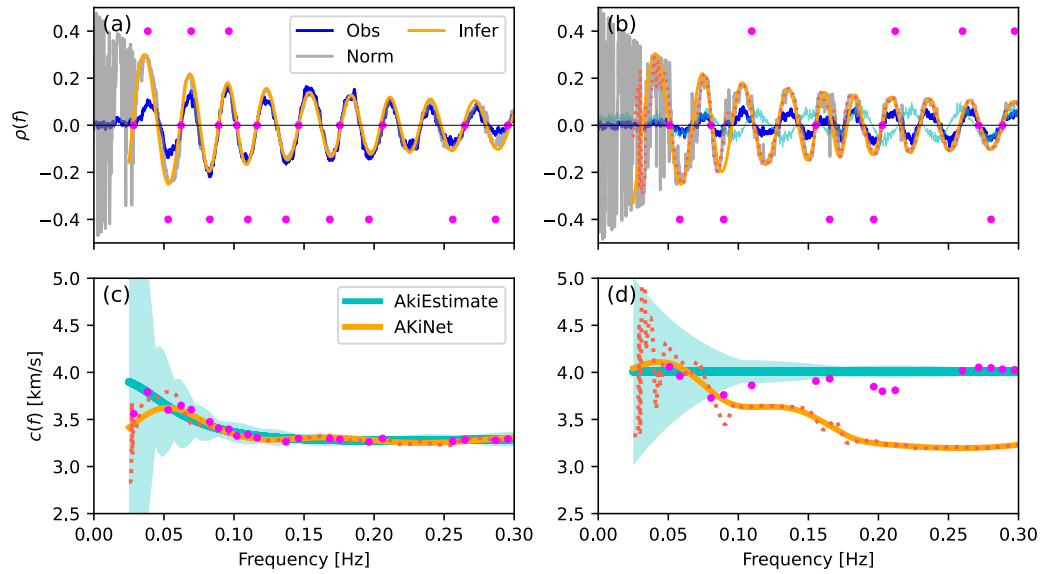


Figure 3. Phase velocity inversion results comparing Rayleigh waves (left column) and Love waves (right column) for station pair XD-RUNG_XD-MTAN ($r = 110$ km). (a, b) Show the noise correlation function (NCF) spectra. The raw, observed NCF (ρ^{obs}) is blue, while the normalized NCF (ρ^{norm}) used for the inversion is gray. The final predicted NCF from AkiNet (ρ^{infer}) is shown as a solid orange line. For comparison, the prediction from AkiEstimate is the cyan line. The pink dots indicate the features (peaks, troughs, and zero-crossings) that AkiEstimate uses to generate its initial model. (c, d) Compare the inverted phase velocity curves ($c^{\text{ivt}}(f)$). The final solution from AkiNet is the thick orange line and the solution from AkiEstimate is the cyan line (with its 95% confidence interval). To illustrate the importance of regularization, the dashed orange line shows an AkiNet result produced without the smoothness constraint ($\{\lambda_1, \lambda_2\} = \{1.0, 1 \times 10^{-20}\}$), demonstrating how the model can overfit to noise. The corresponding NCF prediction for this overfitted model is shown as a dashed line in panel (b). Details on the NCF computation for this data set can be found in Olugboji and Xue (2022).

3.3.2. The Physics-Informed Loss Function

The second component is the loss function, which embeds the governing physics of stochastic wave propagation directly into the training objective (Aki, 1957; Haney et al., 2012). The total loss, L_{total} , is a weighted sum of a data misfit term and a smoothing regularization term:

$$L_{\text{total}} = \lambda_1 L_{\text{Bessel}} + \lambda_2 L_{\text{Smooth}}. \quad (6)$$

The first term, L_{Bessel} , is the primary data misfit component. It uses forward models from Equation 2 or Equation 3 to compute the predicted NCF, ρ^{infer} , from the network's current $c^{\text{ivt}}(f)$. It then quantifies the mean-squared error (MSE) between this prediction and the pre-processed, normalized data, ρ^{norm} . The second term, L_{Smooth} , is a regularization term that enforces physical realism by penalizing roughness in the solution. It is calculated as the MSE of the second derivative of $c^{\text{ivt}}(f)$. The weights, $\{\lambda_1, \lambda_2\}$, are critical for balancing the data fit with the smoothness constraint, especially for noisy data. If λ_2 is too small, the network may overfit noise in the NCF, resulting in a non-physical, oscillatory dispersion curve, as demonstrated in Figure 3. The two loss terms are on fundamentally different numerical scales. The data-fidelity term, L_{Bessel} , measures waveform misfit, while the smoothness term, L_{Smooth} , measures the curvature of the dispersion curve via its second derivative. Since physically plausible dispersion curves are expected to be smooth, the magnitude of this curvature term is naturally much smaller than the data misfit. The small value of λ_2 is therefore chosen to correctly balance these scales, ensuring the solution is primarily driven by the data while being regularized against unphysical roughness.

3.3.3. Training and Optimization

The final component is the training process, where the network's weights and biases are optimized to minimize L_{total} . For this task, we employ the Limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) algorithm. L-BFGS is a quasi-Newton, second-order optimization method that efficiently approximates the Hessian matrix, using information about the loss function's curvature to achieve much faster convergence than first-order methods

like Adam for this class of problems. While L-BFGS can be memory-intensive, the small size of AkiNet makes this a non-issue. Based on our data, we find that the loss weights of $\{\lambda_1, \lambda_2\} = \{1.0, 1 \times 10^{-8}\}$ provide robust results. We use these fixed values for all experiments presented in this work. To visualize how these components contribute to the optimization, we provide an example of the loss evolution during training in Figure S2 in Supporting Information S1. This figure illustrates how the network balances data fit against smoothness and confirms that the final solution is dominated by the data-fidelity term. With this configuration, the entire inversion process for both Love and Rayleigh waves for the 68 station pairs in Figure 5 took less than 25 min on a standard laptop (Apple M1 Pro).

3.4. Comparison With Other Inversion Methodologies

AkiNet shares the same fundamental goal as traditional inversion algorithms like AkiEstimate (Hawkins & Sambridge, 2019) and Matnoise (Menke & Jin, 2015): to invert the NCF for a phase velocity curve, $c(f)$, by minimizing a loss function that balances data misfit and physical realism. However, the implementation strategy differs significantly. Traditional methods typically follow a multi-step workflow: (a) extracting features like zero-crossings from the NCF, (b) generating an initial dispersion curve based on these features and a reference Earth model, and (c) iteratively updating this model to minimize the loss. In contrast, AkiNet performs the inversion in a single, end-to-end process. It starts from a random state and does not require a preliminary feature-extraction step or an initial model guess. Physical constraints are instead embedded directly into the network architecture and loss function—the final linear transformation layer ensures the solution is bounded within a plausible range, while the L_{Smooth} term enforces smoothness.

Second, a more fundamental distinction lies in how the optimization gradients are computed. Traditional algorithms rely on manually derived analytical gradients. For example, AkiEstimate uses the adjoint-state method to efficiently compute the partial derivatives of the forward problem. AkiNet, by virtue of its implementation in a deep learning framework, leverages automatic differentiation. This technique calculates exact gradients by systematically applying the chain rule during the backpropagation step of training. The advantage is twofold: it greatly simplifies the implementation by obviating the need to derive and code complex adjoint solutions, and it provides greater flexibility to modify the physics-informed loss function.

Finally, AkiNet's methodology is fundamentally different from supervised deep learning approaches (Dong et al., 2021; Zhang et al., 2020). These models typically reframe the problem as image segmentation. They convert NCFs into 2D FTAN images and train a deep convolutional network (e.g., a U-Net) to perform a pixel-wise segmentation of the image to “pick” the dispersion curve. This supervised paradigm requires an extensive training process, often involving pre-training on millions of synthetic examples followed by fine-tuning on real data with human-provided labels. In contrast, AkiNet is not an image-based pattern recognizer but an unsupervised, physics-driven inversion tool. It operates directly on the 1D NCF waveform and uses the governing physical equations (Equations 2 and 3) to solve for the underlying parameters, requiring no pre-training or labeled data. In all our benchmarks, we compare AkiNet with AkiEstimate, because its software and methodology are documented, open source, and have been applied to a large NCF data set (Olugboji & Xue, 2022).

3.5. AkiNet Uncertainty Quantification and Future Directions

Quantifying the uncertainty of an inverted model is crucial for making it a trustworthy scientific tool. While traditional inverse methods often estimate parameter uncertainties using linearized approximations of posterior covariance, uncertainty quantification for deep learning models remains an active and challenging area of research (Gawlikowski et al., 2023). The common approaches each present practical difficulties for the current AkiNet framework. Ensemble and bootstrapping methods, while robust, are computationally prohibitive for large-scale applications, as they would require running the AkiNet inversion multiple times for every station pair. Converting AkiNet into a fully Bayesian neural network is a substantial undertaking that would fundamentally alter the method, while techniques like stochastic dropout are incompatible with the L-BFGS optimizer we use for its rapid convergence (Bollapragada et al., 2018).

As a first step toward UQ for AkiNet, we explore the Laplace approximation, a method where the inverse of the loss function's Hessian matrix is used to approximate the posterior covariance of the network's parameters (Kristiadi et al., 2021). Given AkiNet's relatively simple structure, explicitly computing the Hessian is computationally feasible. However, the validity of this approximation for our specific inverse problem faces several

challenges: (a) the noise in the pre-processed NCF, ρ^{norm} , is likely non-Gaussian; (b) the L_{Smooth} regularization term in the loss function complicates the statistical interpretation of the posterior; and (c) the loss is calculated from a nonlinear mapping of the target parameter, $c^{\text{ivt}}(f)$.

A detailed derivation of this method and a discussion of its application are provided in Text S4 in Supporting Information S1. We present this analysis not as a final solution, but as a preliminary investigation into this complex topic. Developing a robust and computationally efficient UQ method for this type of physics-informed inversion remains a key direction for future work.

4. Inversion Results

4.1. AkiNet Performance on High- and Low-Quality NCFs

To demonstrate AkiNet's performance on real data, we first examine a representative station pair that features two distinct signals: a high-quality Rayleigh wave NCF and a more challenging, low-quality Love wave NCF (Figure 3). This example allows us to both validate AkiNet against traditional methods on clean data and to demonstrate its robustness in the presence of noise. For the high-quality Rayleigh wave NCF (Figure 3, left panels), the inversion results of AkiNet and AkiEstimate are in excellent agreement. Both methods produce nearly identical dispersion curves that fit the observed data well. This result is crucial because it confirms that, on clean and unambiguous signals, our PINN approach correctly converges to the same trusted solution as established physics-based algorithms.

The advantage of the AkiNet framework becomes apparent when dealing with the noisy low-amplitude Love wave NCF (Figure 3, right panels). Our amplitude normalization pre-processing step effectively corrects the signal, creating a well-posed waveform for the following inversion (Figure 3b). Here, the resulting dispersion curves from the two methods diverge significantly. The AkiEstimate result suffers from a cycle skip; its inferred waveform is misaligned with the data because the inversion has locked onto the wrong phase oscillation. In contrast, AkiNet's inferred waveform is a visibly better fit to the normalized data across much of the frequency band.

This example highlights a key practical advantage of AkiNet's methodology. The cycle skip in the AkiEstimate result is a classic symptom of its multi-step workflow becoming trapped in a local minimum. Its initial model, derived from picking potentially ambiguous peaks and troughs in the noisy, low-frequency part of the signal, was likely incorrect. AkiNet avoids this pitfall by performing a single end-to-end optimization on the entire normalized waveform. Guided by the smoothness constraint, AkiNet is not dependent on a potentially flawed initial model and is able to find a more globally optimal and physically plausible solution.

Furthermore, our preliminary uncertainty analysis, detailed in Figures S4 and S5 in Supporting Information S1, reinforces this conclusion. The Rayleigh wave inversion results in a cleaner data fit compared to the Love wave inversion (compare Figures S4a and S4b in Supporting Information S1). This difference in data quality is correctly captured by our uncertainty estimates: the confidence intervals for the Love wave dispersion curve are significantly larger than for the Rayleigh wave (compare Figures S4d, S5d and S4c, S5c in Supporting Information S1). This demonstrates that AkiNet's uncertainty quantification can reliably reflect the quality of the input data, assigning higher uncertainty to inversions constrained by noisier NCFs.

4.2. Stress-Testing AkiNet: The Role of Architectural Regularization

We now stress test both methods on more challenging NCFs from longer inter-station paths, which are more oscillatory and often have lower signal-to-noise ratios (Figure 4). These examples highlight how the different algorithmic designs behave when the inverse problem becomes more severely ill-posed.

The Rayleigh wave example (Figure 4, left panels) presents a case where both methods struggle to produce a fully satisfactory result. The AkiEstimate inversion yields a smooth but physically unrealistic dispersion curve with velocities that are too high, suggesting that it failed to capture the correct waveform structure. In contrast, AkiNet successfully fits the data at frequencies below 0.2 Hz but is clearly misled by data contamination at higher frequencies, causing an unphysical, sharp drop in velocity. While neither solution is perfect, AkiNet correctly identifies the structure in the uncontaminated part of the signal. This suggests that its performance could be further

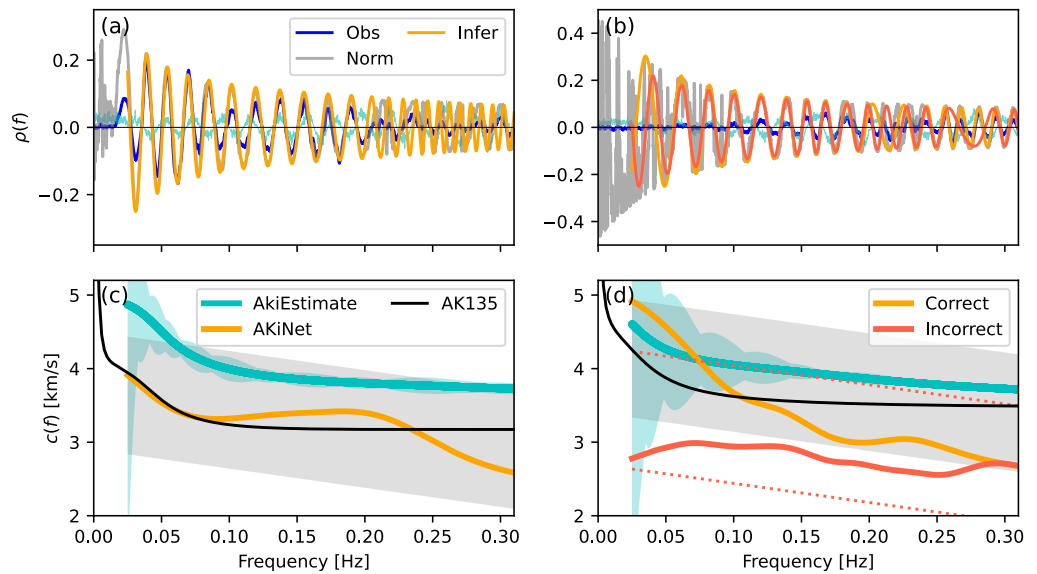


Figure 4. Challenging inversion examples illustrating failure modes and the critical role of architectural regularization. The left column displays the Rayleigh wave inversion for 6A-WP22_6A-WP11 ($r = 200$ km), and the right column shows the Love wave inversion for 1C-TOR1_AF-NBI ($r = 150$ km). The color scheme is consistent with Figure 3. (a, b) Show the noise correlation function spectra. (c, d) Show the corresponding inverted phase velocity curves ($c^{\text{inv}}(f)$). Panel (d) demonstrates the impact of regularization on an ill-posed Love wave problem. The solid orange line is the standard AkiNet solution using the final transformation layer defined in the Methods section. The range of the standard transformation is illustrated by the gray shading. For comparison, the darker orange line is obtained from an altered and less-physical transformation layer. The range of the altered transformation is indicated by the dark orange dotted lines. The AK135 velocity model a good generic reference model to a wide range of seismic phases (Kennett et al., 1995); we use it as a sanity check here. See Figure S3 in Supporting Information S1 for solutions using a specialized loss.

improved by incorporating frequency-dependent data uncertainties to downweight noisy sections of the NCF during inversion.

The Love wave example (Figure 4, right panels) demonstrates an ill-posed problem where the NCF has an extremely low signal-to-noise ratio, offering very little information to constrain the inversion. In such scenarios, the solution becomes highly dependent on regularization. This dependency is powerfully illustrated by the two different AkiNet results shown. The accepted solution (solid orange line) is produced using our standard architectural constraint in the final layer. The alternative, cycle-skipped solution is obtained by altering this constraint.

For $f < 0.2$ Hz, both AkiNet solutions fit the noisy ρ^{obs} almost equally well, thus the prior physical knowledge encoded in the network's final layer acts as a critical tie-breaker. It guides the optimizer toward the more plausible result, showcasing the practical value of embedding physics directly into the network architecture. This approach points toward future work in designing even more sophisticated physical constraints. As a demonstration of this flexibility, we show that for particularly difficult cases, the loss function itself can be augmented. Adding targeted boundary and monotonicity constraints can further guide the inversion toward a stable solution (Figure S3 in Supporting Information S1).

4.3. Regional-Scale Application on the Ugandan Craton

To assess the practical performance of AkiNet at a larger scale, we applied both AkiNet and AkiEstimate to a data set of 68 station pairs located on the Ugandan Craton (Figure 5). This region is understood to be a geologically homogenous crustal block, meaning the underlying phase velocity structure should be statistically similar across all paths (Olugboji et al., 2024). This provides an ideal test bed to evaluate the statistical reliability and failure rate of each inversion method; a more robust method should produce a tighter cluster of results with fewer outliers.

The results show that for relatively clean Rayleigh waves (Figures 5a and 5b), both methods perform well. The bulk of the successful inversions is tightly clustered around a similar mean velocity, consistent with the region's

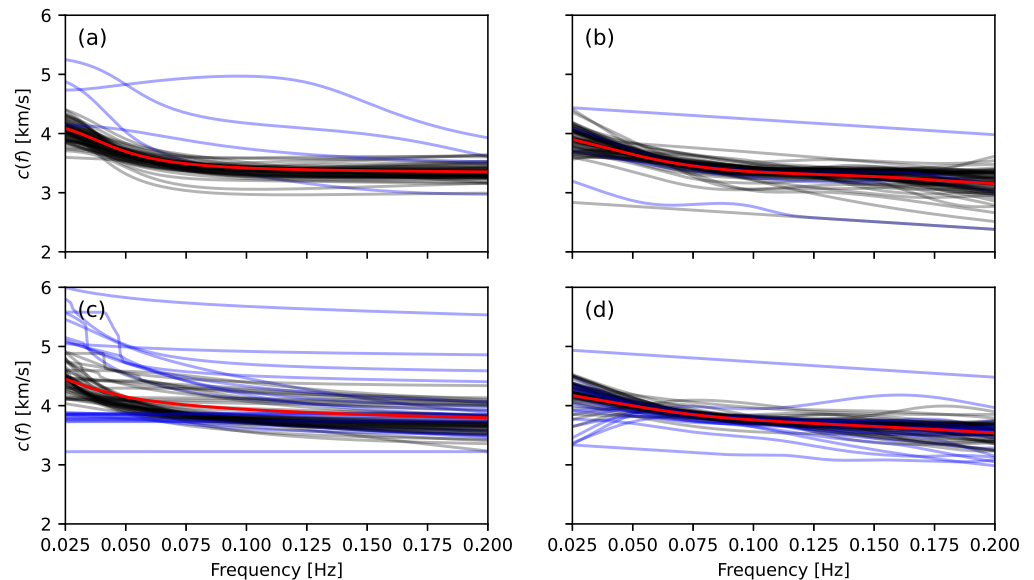


Figure 5. A regional-scale comparison of phase velocity inversions from (a, c) AkiEstimate and (b, d) AkiNet for 68 station pairs on the Ugandan Craton (0° – 5° N, 30° – 35° E). (a, b) Show the Rayleigh wave results. (c, d) Show the Love wave results. Each panel displays the individual dispersion curves, with the calculated mean shown as a solid red line. To highlight differences in reliability, failed or physically unrealistic inversions (outliers) are colored blue. The results visually demonstrate that AkiNet yields significantly fewer outliers than AkiEstimate, particularly for the more challenging Love waves ((d) vs. (c)), indicating a higher success rate at a regional scale.

homogeneity. The primary distinction between the methods emerges in the more challenging Love wave inversions (Figures 5c and 5d). While the core of successful results from both methods remains tightly clustered, AkiEstimate produces a significantly larger number of outliers, indicating a higher failure rate on this more difficult data type. In contrast, AkiNet yields a much more consistent set of results with visibly fewer failures, demonstrating its robustness at scale.

Furthermore, these failures between the two methods are fundamentally different. The outliers from AkiEstimate are often physically unrealistic flat lines, a behavior consistent with the cycle-skipping local minima identified in our single-station examples. When AkiNet fails, however, it exhibits a predictable and diagnostic behavior: it produces a straight, sloped line. This occurs when the NCF quality is too poor to inform the loss function, causing the network's output to default to the baseline reference model defined by its architectural regularization. The lower failure rate, combined with this diagnostic failure mode, makes the AkiNet data set easier to quality control and demonstrates the practical reliability of its end-to-end, physics-constrained approach (examples of these failed inversions are provided in Supporting Information S1).

4.4. AkiNet Generalization and Performance on a Diverse Data Set

To address the critical questions of generalizability and computational feasibility, we conducted a large-scale benchmark test of AkiNet. We applied our method to 539 NCFs sampled from the four distinct tectonic domains across Africa, as shown in Figure 1. This diverse data set, encompassing stable cratons (C1), mobile belts (C2), Pan-African terranes (C3), and rifts (C4), provides a robust test of AkiNet's performance under a wide range of real-world signal and noise conditions.

First, to evaluate the robustness of our inversion results, we compare the distributions of the inverted phase velocities from AkiNet with those from the AkiEstimate benchmark. As shown in Figure 6, AkiNet consistently produces tighter, more stable distributions of phase velocities across all tectonic domains, for both Rayleigh and Love waves. For the more challenging Love wave data, AkiEstimate produces a significantly larger number of outliers, indicating a higher failure rate on this more difficult data type. In contrast, AkiNet yields a much more consistent set of results with visibly fewer failures. This result is a key demonstration of AkiNet's practical feasibility and robustness at scale.

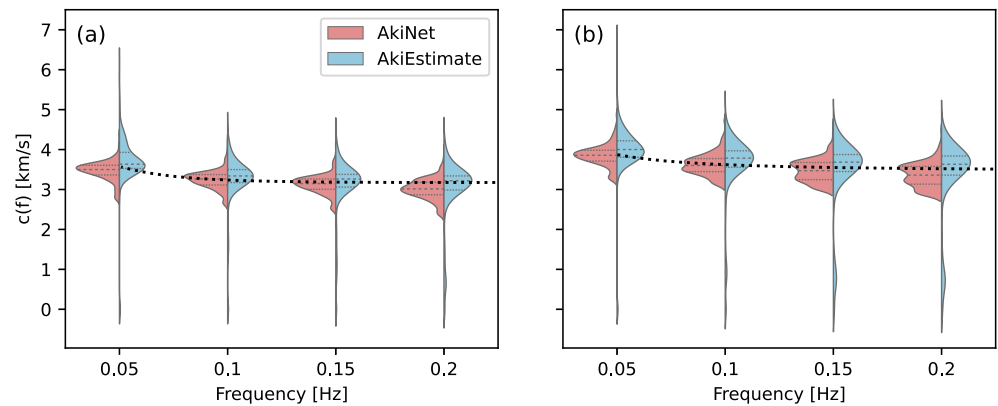


Figure 6. Comparison of AkiNet and AkiEstimate performance on 539 noise correlation functions, randomly sampled from four distinct tectonic domains (C1–C4) in Africa. (a) Violin plots showing the distribution of inverted Rayleigh wave phase velocities at representative frequencies (e.g., 0.1 Hz). Dashed lines inside each half violin show the quartiles of the distribution. (b) Corresponding distributions for Love wave velocities. The results demonstrate that AkiNet (pink) produces more stable distributions with fewer outliers compared to AkiEstimate (blue), particularly for the noisier Love wave data. The black dotted lines show the phase velocities for the AK135 reference Earth model (Kennett et al., 1995), provided for context.

While AkiNet must be trained independently for each NCF, its lightweight architecture and end-to-end nature make it highly efficient. Furthermore, since each inversion is an independent process, the workflow is trivially parallelizable, making AkiNet a computationally practical and scalable solution for large-scale ambient noise studies.

5. Discussion

This work introduces AkiNet, a PINN designed to reliably and efficiently invert ambient noise cross-correlations for surface wave dispersion. Our results demonstrate that AkiNet performs as well as, or better than, established inversion algorithms. Its primary practical advantage lies in streamlining the complex multi-step workflow, which is typical of traditional methods, into a single end-to-end process. Researchers no longer need to manage separate codes and ad-hoc files for feature picking, initial model generation, and final inversion. Furthermore, AkiNet's robustness, evidenced by the significantly lower rate of outlier results for challenging Love waves, and its predictable, diagnostic failure mode make it particularly well-suited for the large-scale automated processing required in modern seismology.

While our results demonstrate AkiNet's robustness, it is important to acknowledge the assumptions underlying the method. The forward model relies on the SPAC theory, which assumes a diffuse, isotropic noise wavefield (Tsai, 2011; Tsai & Moschetti, 2010). When these ideal conditions are violated, the resulting NCFs can become noisy and complex (Stehly et al., 2006; Zhan et al., 2013). AkiNet's end-to-end, full-waveform fitting approach is designed to be robust to precisely these non-ideal NCFs, finding the best-fitting physical model that explains the entire observed waveform. This direct waveform-fitting approach also bypasses the significant overhead of supervised CNNs, which rely on larger, more complex architectures that require longer pre-training on massive (and often unavailable) labeled data sets of 2D images generated from NCFs. Beyond the underlying physics, AkiNet's effectiveness is also guided by its architectural priors, namely the bounding corridor and smoothness constraint. To quantify their influence, we conducted sensitivity tests. These tests confirm that our choice for the bounding corridor is a deliberately generous hyperparameter that provides essential regularization without being overly restrictive; the final solution is robustly driven by the data-fidelity term. Having established the robustness of the current framework, several avenues for future methodological improvement remain. The network's design can be enhanced by embedding more sophisticated physics, for instance, by incorporating frequency-dependent data uncertainties into the loss function or by expanding the forward model to account for multi-mode dispersion (Hu et al., 2020; Wang et al., 2019).

Ultimately, the development of AkiNet is motivated by the long-standing goal of building a global, high-resolution surface wave model from ambient noise. As stated in our introduction, achieving this has been hindered by two fundamental obstacles: the computational cost of generating NCFs globally and the difficulty of reliably inverting the resulting massive, heterogeneous-quality data set. While complementary work is underway to address the first challenge, AkiNet provides a robust and scalable solution to the second. By automating the reliable extraction of dispersion information, AkiNet removes a critical bottleneck. This positions the community to finally construct a global ambient noise dispersion database, aspiring to do for ambient noise what seismologists have done for earthquake-derived surface waves (Ekström, 2011; Ekström et al., 1997). Such a resource would be invaluable for refining our next-generation models of Earth's lithosphere.

6. Conclusion

In this study, we developed and validated AkiNet, a PINN framework that provides a robust solution to the nonlinear inverse problem of surface wave dispersion from NCFs. Our results progressively demonstrate AkiNet's practical advantages for this task. We first show its ability to avoid cycle-skipping errors that can trap traditional methods in local minima. We then illustrated how its architecturally embedded physical constraints allow it to navigate severely ill-posed problems on noisy, long-path data where other methods fail. Finally, our regional-scale application on the Ugandan Craton confirmed its reliability, showing a significantly lower failure rate than traditional algorithms. Each of these findings validates AkiNet as a practical and scalable tool. By overcoming key methodological bottlenecks in robustness and automation, this work represents a critical step toward the construction of the first global surface wave dispersion model derived entirely from ambient noise, a long-standing goal in modern seismology.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

AkiNet codes and the data used to generate all figures can be found on Zenodo (Xue, 2025). NCF and AkiEstimate data are taken from our previous publication (Olugboji & Xue, 2022).

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